

SUGANYA P

GenAI Systems Engineer | AWS Bedrock | LLM Systems | RAG & Orchestration
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PROFESSIONAL SUMMARY

GenAI Engineer with 8+ years of experience across QA automation, backend systems, and AI-based application development. Hands-on experience in building RAG workflows, LangGraph orchestration systems, AI validation pipelines, and governance-aware GenAI applications using Python and AWS.

Worked on end-to-end AI systems involving semantic retrieval, structured extraction, observability, evaluation workflows, and controlled LLM execution. Strong background in system reliability, workflow validation, API testing, and modular backend development.

CORE SKILLS

GenAI & AI Engineering : LangGraph , RAG Workflows, Prompt Engineering, OpenAI APIs, AWS Bedrock, Hugging Face, Semantic Retrieval , Vector Search, FAISS

Backend & Infrastructure : Python, Flask, AWS Lambda, API Gateway, CloudWatch

Evaluation & Observability : LangSmith, RAGAS, DeepEval, Structured Logging, AI Validation, Workflow Monitoring

AI/ML Libraries : PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy

Governance & Evaluation — PII Masking, Prompt Firewall, Validation Pipelines, Responsible AI, Observability

Architecture & Reliability — System Design, Logging, Traceability, Routing Logic, Performance Monitoring, Debugging

PROFESSIONAL EXPERIENCE

Independent GenAI Engineer | Remote | Feb 2024 – Present

Agentic-RAG-Resume-Intelligence

Built a governance-aware AI resume intelligence platform using LangGraph, Flask, FAISS, and Sentence Transformers to move beyond traditional keyword-based ATS systems through contextual semantic retrieval and deterministic evaluation workflows.

- Developed modular LangGraph orchestration pipelines for parsing, extraction, retrieval, analysis, and evaluation. Developed PDF parsing and competency extraction pipelines.
- Replaced keyword ATS matching with semantic competency-based retrieval using Sentence Transformers and FAISS.
- Implemented weighted semantic scoring for realistic recruiter-style candidate evaluation
- Added governance controls including PII masking, workflow validation, and safety filtering
- Integrated evaluation frameworks like LangSmith, RAGAS, and DeepEval for retrieval quality and scoring validation

- Designed explainable recruiter-facing outputs including strong matches, missing skills, retrieval precision, and score consistency

Technologies : Python, Flask, LangGraph, LangChain, FAISS, Sentence Transformers, LangSmith, RAGAS, DeepEval

AI Companion System for Guided Conversations – (Bro-Buddy)

Production-oriented GenAI workflow built using AWS Bedrock and serverless architecture with focus on reliability, observability, and controlled execution.

- Designed and deployed a serverless GenAI system using AWS Bedrock, Lambda, and API Gateway
- Built deterministic routing workflows reducing unnecessary LLM calls by 30–40% and improving latency
- Implemented multi-layer safety pipeline with PII masking, prompt injection filtering, validation, and fallback handling
- Developed orchestration flow integrating routing, memory handling, validation, and response control
- Built structured observability for latency tracking, routing decisions, token monitoring, and debugging
- Achieved 92% routing accuracy with controlled AI execution and governance-focused workflow design

Technologies : AWS Bedrock, Lambda, API Gateway, Python, Flask, CloudWatch, Docker

AI-Powered Book Recommendation System – (Kittylit)

AI-based children's book recommendation platform using orchestration workflows, RAG pipelines, and governance-first AI design..

- Built orchestration workflows for recommendation routing and fallback handling
- Developed RAG-based semantic retrieval pipelines using FAISS and Sentence Transformers
- Implemented cache → DB → API recommendation architecture
- Added governance guardrails, safety filters, and explainability logging.
- Designed modular backend services using Flask and Python.
- Built conversational recommendation support workflows

Technologies : Python, Flask, RAG, FAISS, Sentence Transformers, HTML, CSS

AI Systems Reliability Engineer (Contract) | Jan 2024 – Feb 2024

- Monitored AI system performance using CloudWatch and structured logging
- Supported backend debugging and routing issue analysis for workflow stability
- Improved operational reliability through structured validation and issue tracking
- Collaborated with backend and AI teams to analyze response inconsistencies and deployment issues

Earlier Experience | GE Healthcare • Infosys • Caterpillar | 2013 – 2019

- Worked on enterprise system validation, QA automation, and defect reduction initiatives
- Improved production stability through structured testing and regression workflows
- Contributed to system reliability, validation processes, and release quality improvements

PUBLICATIONS & COMMUNITY CONTRIBUTIONS

- Authored technical content on GenAI system design, governance, evaluation, and cost-aware AI architecture
- Developed practical frameworks for AI orchestration, automation vs agents, and responsible AI workflows
- Proposed governance-focused approaches for safe and reliable AI deployment

EDUCATION & CERTIFICATIONS

- PG Diploma in AI & ML — NIT Warangal (2023)
- AI Security & Governance Certification — Securiti (2025)
- Building AI Applications with Amazon Bedrock — LinkedIn Learning (2026)
- Executive MBA (IT Management) — ISMS Mumbai (2018)
- B.E. Computer Science — PMU (2013)